Sentiment Analysis Overview

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This lecture was created based on materials from Prof. Pilsung Kang of Seoul National University.

Contents

1. Overview
2. Architecture
3. Lexicon-based Approach
4. Machine Learning-based Approach

Definition

- What is Sentiment Analysis?
  - Computational study of opinions, sentiments, evaluations, attitudes, appraisal, affects, views, emotions, subjectivity, etc., expressed in **text**.
    - Text: Reviews, blogs, discussions, news, comments, feedback,…
  - Sentiment Analysis is sometimes called **Opinion Mining**

Backgrounds

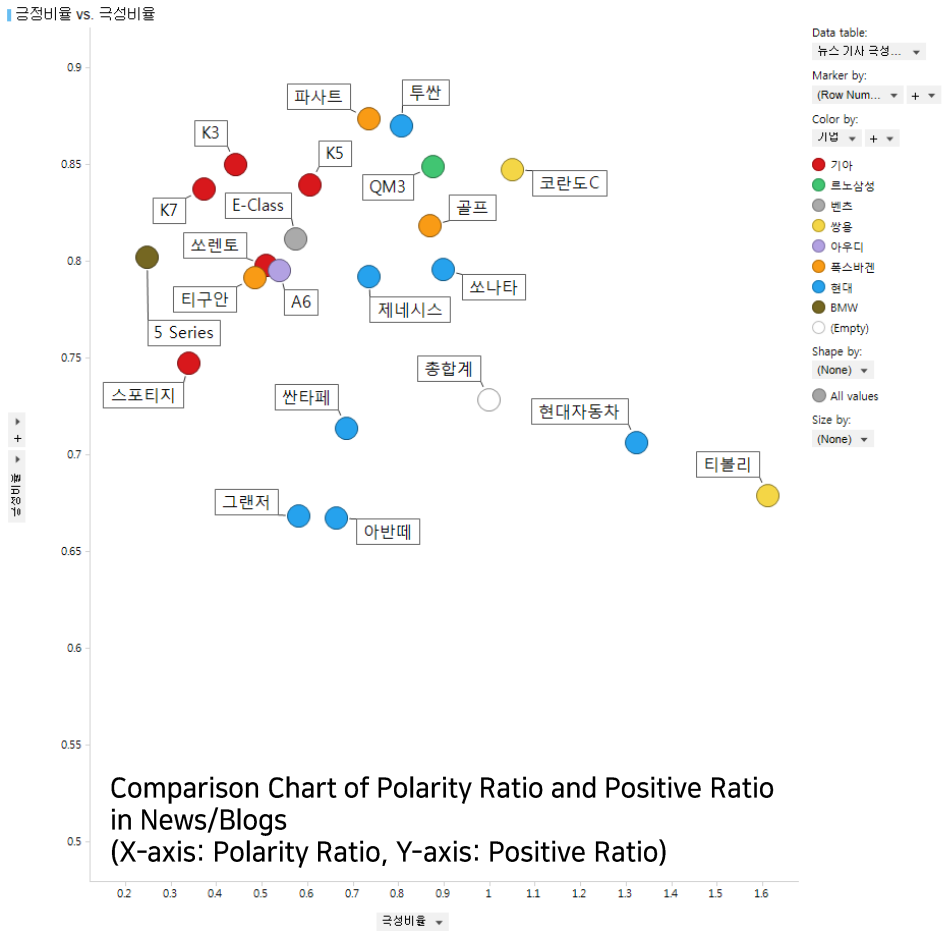
- Typical Usage of Sentiment Analysis
  - Extract from text how people feel about different products
  - Sentiment analysis can be tricky
    - **`Honda Accords and Toyota Camrys are nice sedans`**
    - **`Honda Accords and Toyota Camrys are nice sedans, but hardly the best car on the road`**

Backgrounds

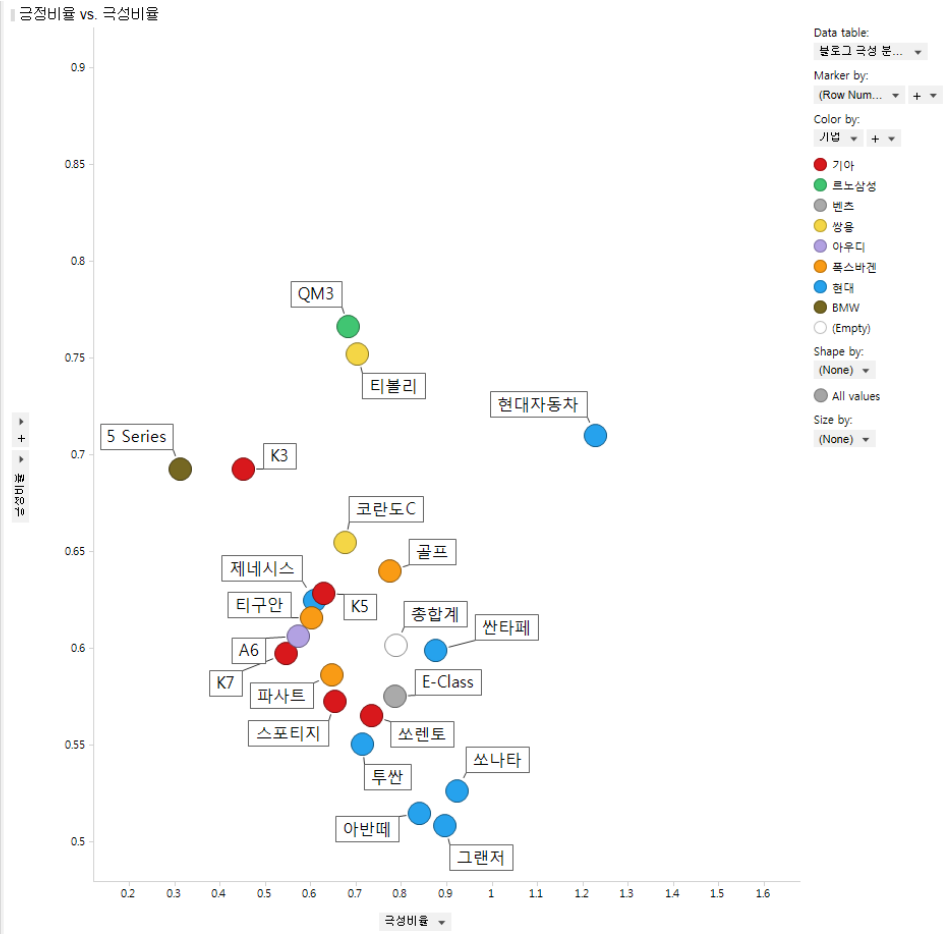
- Opinions are widely stated
  - Organization's internal data
    - Customer feedback from emails, call centers, etc.
  - News and reports
    - Opinions in news articles and commentaries
  - Word-of-mouth on the web
    - Personal experiences and opinions about anything in reviews, forums, blogs, Twitter, micro-blogs, etc.
    - Comments about articles, issues, topics, reviews, etc.
    - Posting at social networking sites (Facebook, Instagram, etc.)

Applications

- Business and organizations
  - Benchmark products and services; market intelligence
    - Business spend a huge amount of money to find customer opinions using consultants, surveys and focus groups, etc.

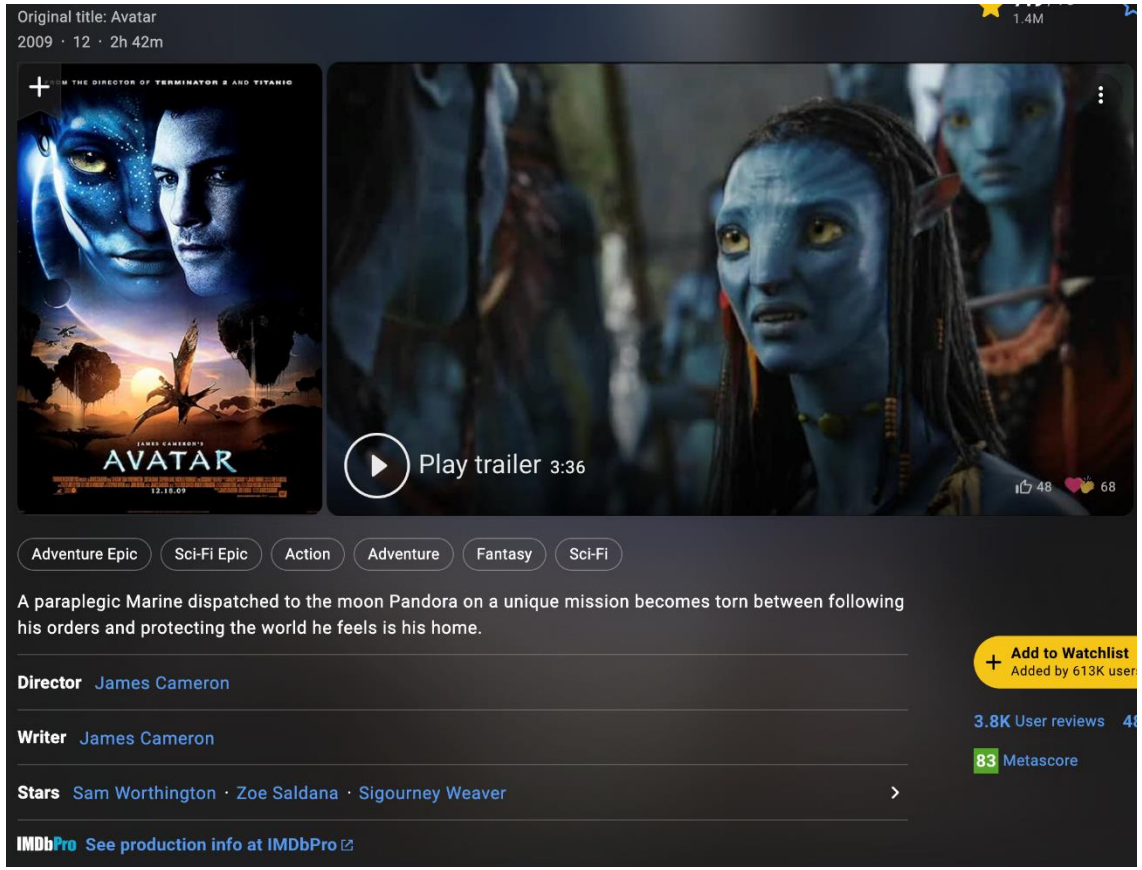


Comparison Chart of Polarity Ratio and Positive Ratio in News/Blogs
(X-axis: Polarity Ratio, Y-axis: Positive Ratio)



Applications

- Individuals
  - Make decisions to purchase products or to use services
  - Find public opinions about political candidates and issues



8/10

After a decade, Cameron sets the avatar of our new generation

Spoiler

Helpful · 1.7K 924

Misagh · Dec 13, 2009 · Permalink

9/10

Cameron Returns

Spoiler

Helpful · 894 529

Samiam3 · Dec 12, 2009 · Permalink

9/10

A fantastic journey.

You know the feeling when you start watching a movie, realizing it's almost 3 hours, and saying to yourself - "Well this is probably gonna be mostly boring", well, the creators of this movie didn't let that happen, rest assures.

From the beginning up until the ending, every scene is perfected, with amazing acting, stunning directing, great visual effects, and astounding sound effects.

The story building is just great , characters are well-developed , and the concept is brilliant.

... only thing that's missing to me, is a fitting soundtrack which could've give some scenes a great

Helpful · 12 2

Applications

- Ad placement
  - Place an ad if one praises a product
  - Place an ad from a competitor if one criticizes a product

Guest

Why sentiment analysis is the future of ad optimization

PETER YARED MARCH 20, 2011 6:21 PM

TAGS: KLOUT, PEERINDEX



[Peter Yared is the vice president of apps at Webtrends, which acquired Transpond, a social-apps developer he founded.]

Sentiment analysis is a hot new trend in social media, with the promise of helping brands understand what consumers are thinking and saying about their products. Products including early contender Radian 6, newcomers such as BuzzLogic, and my own company's Webtrends Social Measurement product are becoming pervasive in marketing organizations. But while consumer sentiment is important, what's much more important is revenue.

SenticNet

Helping machines to learn, leverage, love.

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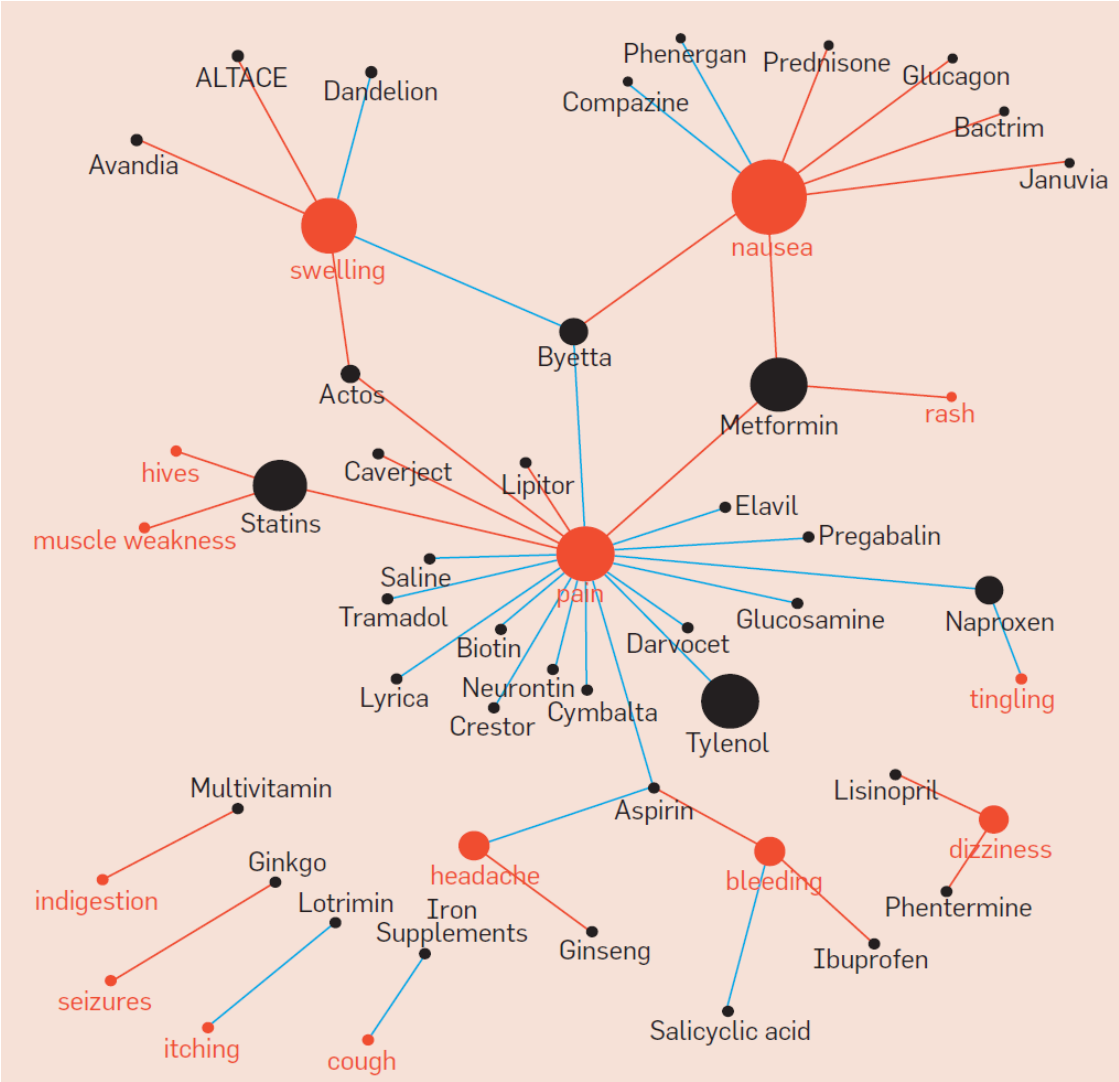
KDD WISDOM

As of 2021, KDD WISDOM has been discontinued: if you are interested in publishing a workshop paper on sentiment analysis, please consider ***ICDM SENTIRE***. WISDOM (Workshop on Issues of Sentiment Discovery and Opinion Mining) aimed to explore how the wisdom of the crowds affects the evolution of the Web and of businesses gravitating around it. In particular, the workshop series explored two different stages of sentiment analysis: the former focusing on the identification of opinionated text over the Web, the latter focusing on the classification of such text either in terms of polarity detection or emotion recognition.

- ***WISDOM'20 (KDD 2020, August 24th, San Diego)***
- ***WISDOM'19 (KDD 2019, August 5th, Anchorage)***
- ***WISDOM'18 (KDD 2018, August 20th, London)***
- ***WISDOM'17 (KDD 2017, August 14th, Halifax)***
- ***WISDOM'16 (KDD 2016, August 14th, San Francisco)***
- ***WISDOM'15 (KDD 2015, August 10th, Sydney)***
- ***WISDOM'14 (KDD 2014, August 24th, New York)***
- ***WISDOM'13 (KDD 2013, August 11th, Chicago)***
- ***WISDOM'12 (KDD 2012, August 12th, Beijing)***

Applications

- Relationships between Drugs and Symptoms



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Problem Statement: Abstraction

- Consists of two parts
  - (1) Opinion definition
    - What is an opinion?
  - (2) Opinion summarization
    - Opinions are subjective
    - An opinion from a single person (unless a VIP) is often not sufficient for action
    - We need opinions from many people, and thus need to summarize those opinions

Problem Definition

Entity and Aspect/Feature

Id: Abc123 on 5-1-2008 *"I bought an iPhone a few days ago. It is such a nice phone. The touch screen is really cool. The voice quality is clear too. It is much better than my old Blackberry, which was a terrible phone and so difficult to type with its tiny keys. However, my mother was mad with me as I did not tell her before I bought the phone. She also thought the phone was too expensive, …"*

- What do we see?
  - Opinion targets: entities and their features/aspects
  - Sentiments: positive and negative
  - Opinion holders: persons who hold the opinions
  - Time: when opinions are expressed

Problem Definition

- Two main types of opinions
  - Regular opinions: Sentiment/opinion expressions on some target entities
    - Direct opinions: "The touch screen is really cool."
    - Indirect opinions: "After taking the drug, my pain has gone."
  - Comparative opinions: Comparisons of more than one entity
    - "iPhone X is better than Galaxy 9"
  - Focus more on regular opinions in this lecture

Problem Definition

- A (Regular) Opinion
  - An opinion has the following basic components

$$(g_i, so_{ijk}, h_j, t_k)$$

where

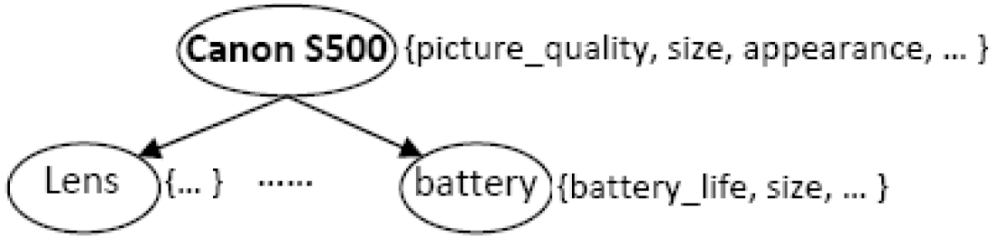
- g_i is a target.
- so_{ijk} is the sentiment value (positive, negative, neutral, or a rating score) of the opinion from opinion holder h_j on target g_i at time t_k.
- h_j is an opinion holder.
- t_k is the time when the opinion is expressed.

Problem Definition

- Opinion Target
  - In some cases, opinion target is a single entity or topic.
    - "I love G5" and "I support tax cut."
  - But in many other cases, it is more complex.
    - "I bought an iPhone a few days ago. It is such a nice phone. The touch screen is really cool."
    - Opinion target of the 3rd sentence is not just touch screen, but the "touch screen of iPhone"
    - "I support tax cut for the middle class, but not for the rich…"

Problem Definition

- Entity and Aspect
  - Definition of Entity: An entity e is a product, person, event, organization, or topic.
  - e is represented as a hierarchy of **components**, **sub-components**, and so on.
    - Each node represents a component and is associated with a set of **attributes** of the component.



    - An opinion can be expressed on any node or attribute of the node.
    - For simplicity, we use the term **aspects (features)** to represent both components and attributes.

Problem Definition

- Definition of an Opinion as a Quintuple

$$(e_i, a_{jl}, so_{ijkl}, h_j, t_k)$$

where

- e_i is a target entity/object
- a_{jl} is an aspect/feature/attribute/facet of the entity e_i
- so_{ijkl} is the sentiment value (positive, negative, neutral, or a rating score) of the opinion from opinion holder (source) h_j on aspect a_{jk} of entity e_i at time t_k
- h_j is an opinion holder
- t_k is the time when the opinion is expressed

Problem Definition

- Quintuple Opinion Examples

> **Id: Abc123 on 5-1-2008** "*I bought an iPhone a few days ago. It is such a nice phone. The touch screen is really cool. The voice quality is clear too. It is much better than my old Blackberry, which was a terrible phone and so difficult to type with its tiny keys. However, my mother was mad with me as I did not tell her before I bought the phone. She also thought the phone was too expensive, …*"

- (iPhone, General, Positive, Abc123, 5-1-2008)
- (iPhone, touch_screen, Positive, Abc123, 5-1-2008)
- …

Problem Definition

Purpose: Structure the unstructured

- Given an opinionated document,
  - Discover all quintuples $(e_i, a_{jl}, so_{ijkl}, h_j, t_k)$
  - Or, solve some simpler form of the problem
    - E.g. Classify the sentiment of the entire document
  - With the quintuples, unstructured texts transformed into structured data.

Rational vs. Emotional Evaluation

- Rational Evaluation
  - Many evaluation/opinion sentences express no emotion
    - The voice of this phone is clear
- Emotional evaluation
    - I love this phone
- Some emotion sentences express no (positive or negative) opinion/sentiment
  - I am so surprised to see you

Abstraction: Opinion Summary

- Feature-based opinion summary

""I bought an iPhone a few days ago. It is such a nice phone. The touch screen is really cool. The voice quality is clear too. It is much better than my old Blackberry, which was a terrible phone and so difficult to type with its tiny keys. However, my mother was mad with me as I did not tell her before I bought the phone. She also thought the phone was too expensive, …"

1.

….

Feature Based Summary of iPhone:

Feature1: **Touch screen**

Positive: 212

- *The touch screen was really cool.*
- *The touch screen was so easy to use and can do amazing things.*

…

Negative: 6

- The screen is easily scratched.
- I have a lot of difficulty in removing finger marks from the touch screen.

…

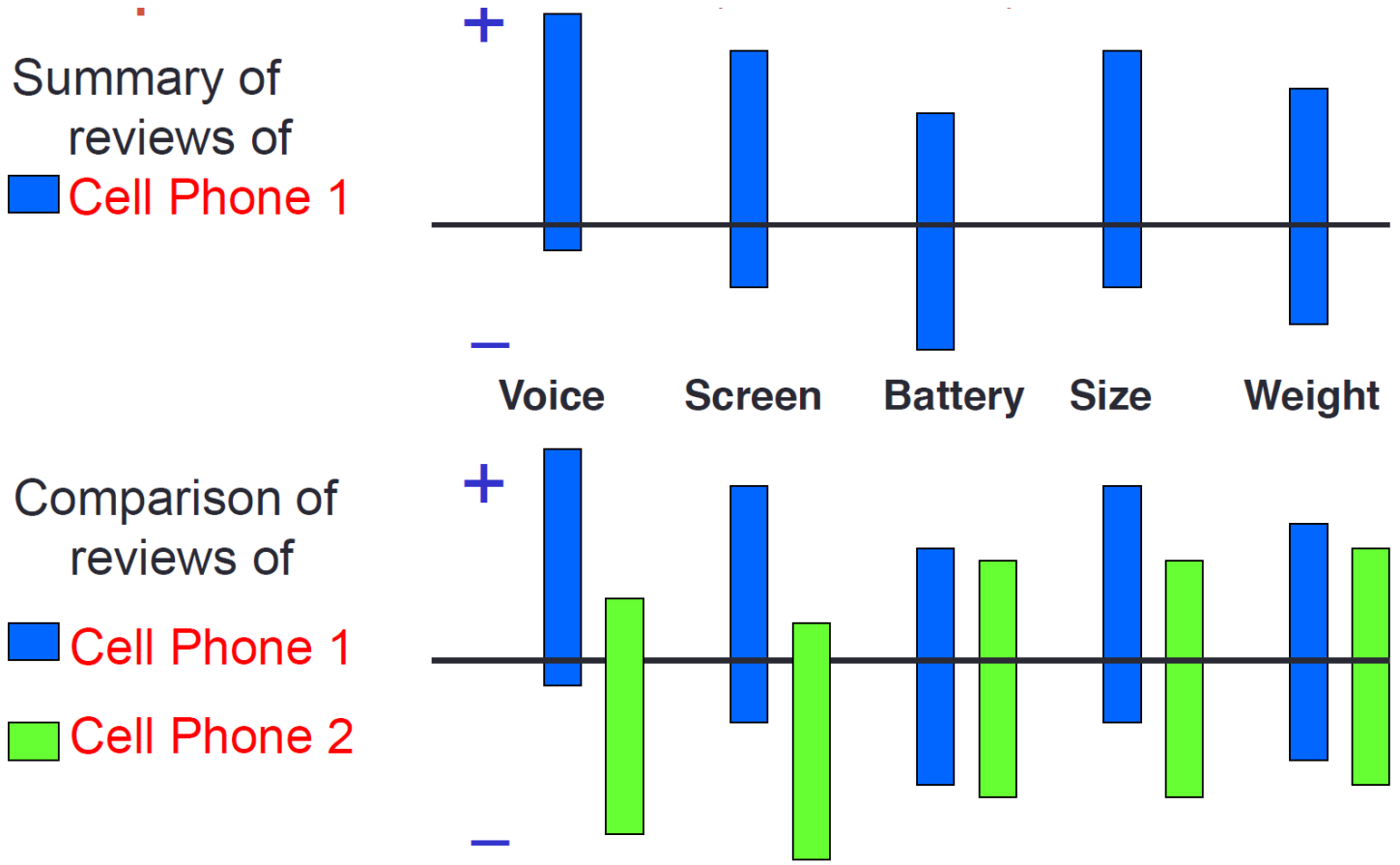
Feature2: **voice quality**

…

Note: We omit opinion holders

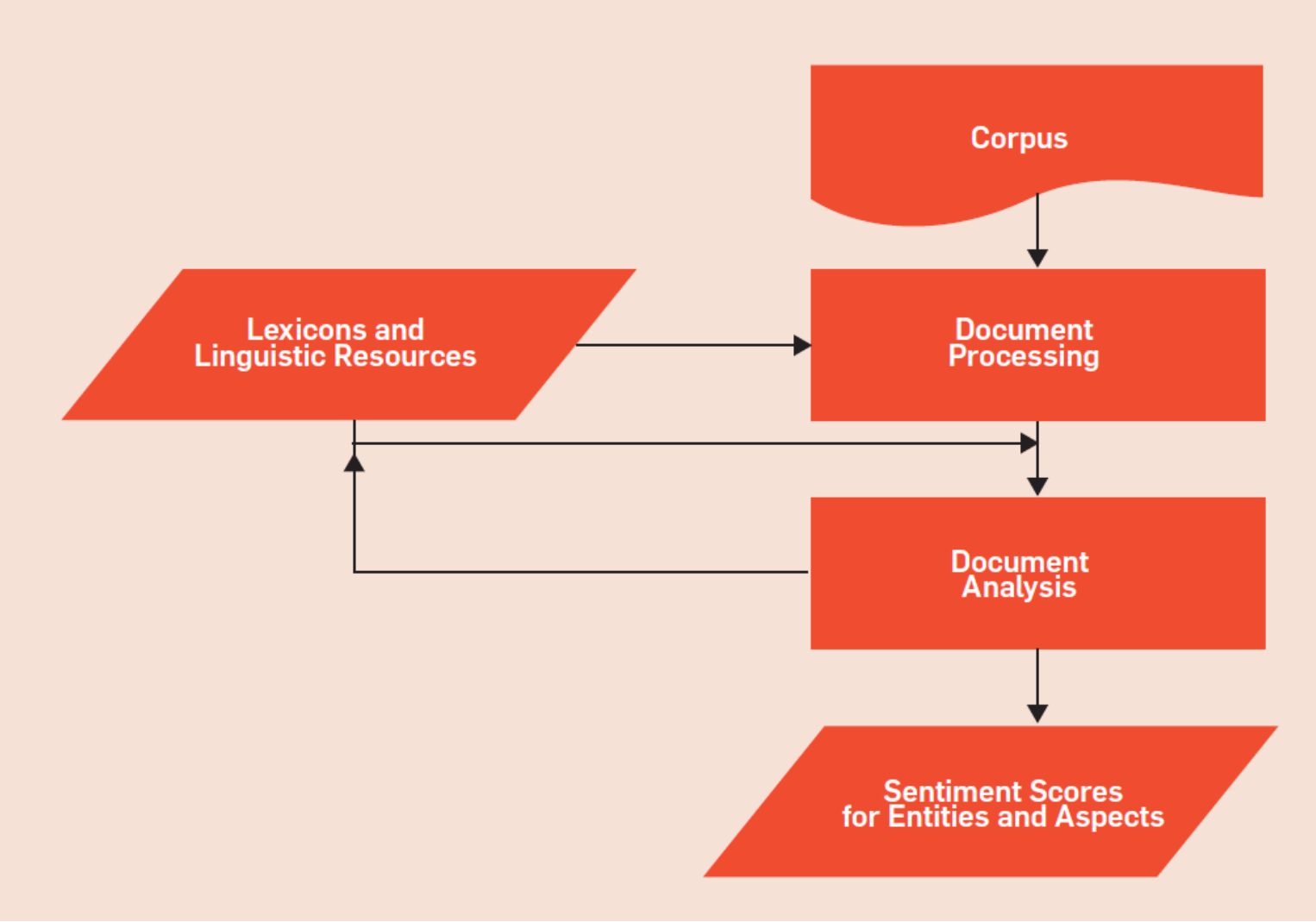
Abstraction: Opinion Summary

- Opinion Comparison



Sentiment Analysis Architecture

- Architecture of generic sentiment analysis system



Document Sentiment Classification

- Document Sentiment Classification
  - Classify a whole opinion document based on the overall sentiment of the opinion holder
    - Classes: positive, negative (possibly neutral)
    - Neutral or no opinion is hard. Most papers ignore it.
  - Goal: discover (_ , _ , so, _, _) where e, a, h, and t are ignored.
    - Reviews usually satisfy the assumption (4/5 stars → positive, 1/2 stars → negative)
    - Many forum postings and blogs do not. They can mention and compare multiple entities or they express no sentiments.
  - Assumption: a document is written by **a single person** and expresses opinion/sentiment on **a single entity**.
  - It is basically a text classification problem

Document Sentiment Classification

- Supervised Learning Formulation
  - Training and test data: movie reviews with star ratings
    - 4-5 stars as positive and 1-2 stars as negative
    - Neutral is ignored
  - Key: feature engineering
    - Typically unigrams (bag of individual words)
    - Term frequency and different IR weighting schemes
    - Part of speech (POS) tags
    - Opinion words and phrases
    - Negations
    - Syntactic dependency
  - Too coarse for most applications

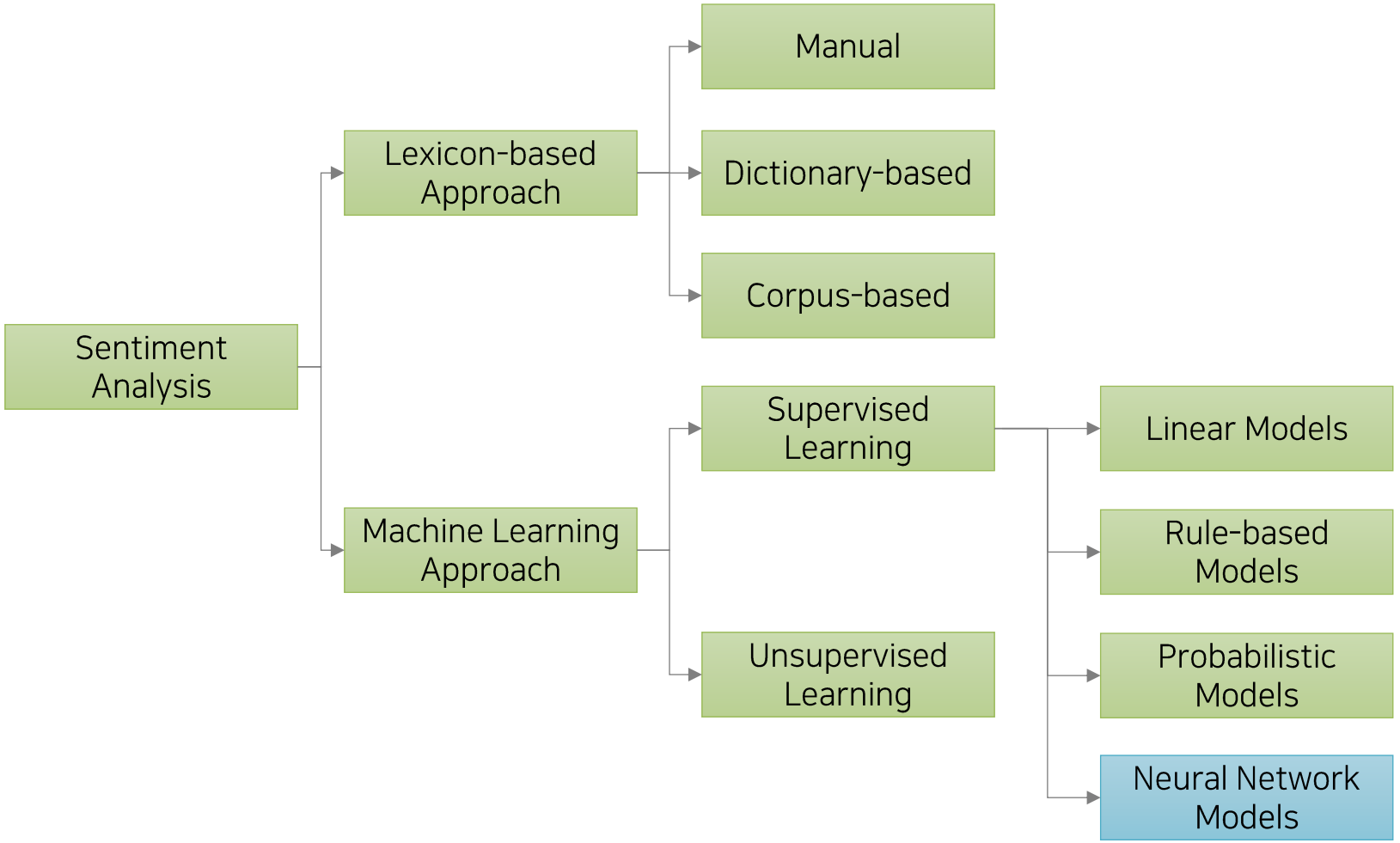
Sentence Sentiment Classification

- Sentence Sentiment Classification
  - Assumes a single sentiment per sentence
  - Not always true, so one can classify clauses instead

- Two steps of sentence sentiment classification
  - Subjectivity classification
    - To identify subjective sentences
  - Sentence classification of subjective sentences
    - As positive or negative

Sentiment Analysis Approaches

- Sentiment classification techniques



Any Questions?